



Zinc battery with biodegradable electrolyte from crab shells

Demand for renewable energy and electric vehicles with a demand for sustainable batteries that can store generated energy and power engines is increasing, but the demand for sustainable batteries is difficult to meet. A paper published in the journal *Matter* has described that scientists have created a zinc battery with a biodegradable electrolyte from crab shells. The batteries utilise electrolyte ions to commute between positively and negatively charged terminals. An electrolyte can be in different forms, such as liquid, paste, or gel, and many batteries use flammable and corrosive chemicals. The new battery, which can store power from large-scale wind and solar sources, uses a gel electrolyte produced from a biological material called chitosan.

Drug delivery systems using graphite nanomaterials

It was difficult to determine if the materials used for drug delivery in the human body are safe or not. In a study published in the journal *Materials Today: Proceedings*, researchers describe the recent development in drug delivery using nanoparticles based on graphite. As they possess unique qualities, such as their 2D structure, mechanical stability, good conductivity, chemical stability, biocompatibility, and vast surface area, graphene and graphene oxide are excellent materials for biological applications. They are more biocompatible and have a higher specific area than other drug delivery systems; graphite based nanomaterials have a higher drug loading rate. For a sophisticated drug delivery system and controlled drug release, graphene based nanomaterials are likely to be used as nanocarriers.

Memristor cells working like artificial synapses

Scientists from Jülich are doing research for more than 15 years on special data storage devices and components that can have similar properties to the synapses in the human brain. They have published a guide for memristor hardware design to understand the various physical and chemical effects in memristors and project the influence of these effects on the switching properties of memristive cells and their reliability. Memristive cells make it possible to process the gigantic data sets directly in memory without transporting them back and forth between processor and memory. They can also be interconnected to form high-density matrices that enable neural networks to learn locally. This edge computing thus shifts computations from the data centre to the factory floor, the vehicle, or the home of people in need of care.

Fully recyclable and biodegradable printed circuits

Electronic devices are built using various materials, some of which are very precious to be dumped thoughtlessly. Researchers at the Department of Energy's Lawrence Berkeley National Laboratory (Berkeley Lab)



A fully recyclable and biodegradable printed circuit (Credit: Marilyn Sargent/Berkeley Lab)

and UC Berkeley have developed a fully recyclable and biodegradable printed circuit to overcome this problem. Researchers have developed a printable conductive ink composed of biodegradable polyester binders, conductive fillers such as silver flakes or carbon black, and commercially available enzyme 'cocktails' made from BC-Lipase. The circuit was recorded as completely degraded within 72 hours, allowing researchers to recover metals without any additional procedures. The technology promises a sustainable alternative to single-use plastics in transient electronic devices.

Robots that can power themselves by sketching electrical circuits

It is necessary to make robots self-reliant to complete missions in different familiar or unfamiliar environments. The robots to operate in the uncertain and remote environment would require their ability to actively acquire electrical power from their surroundings. Researchers at Worcester Polytechnic Institute have introduced a novel robotic system that can power itself by painting electrical circuits using graphene based conductive ink. It can autonomously rearrange the surroundings to obtain the maximum energy from a given power source. These robots can stay functional for long by sketching new circuits and can independently maximise energy in its surroundings.